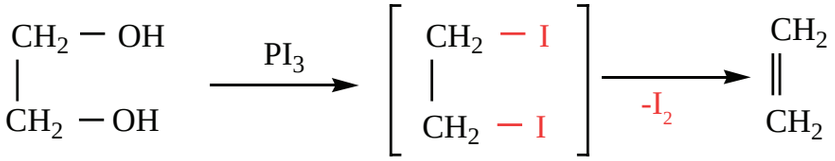
Reactions of Glycol

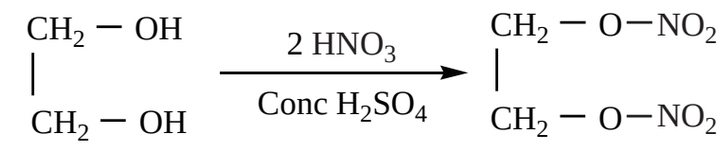
Ethylene glycol contains two primary alcoholic groups and it exhibits the usual reactions of hydroxyl group. Like other primary alcohols, it reacts with metallic sodium to form monosodium glycolate and disodium glycolate. The hydroxyl groups can be converted to the halide groups by treating glycol with halic acid (or with PCl_5 / PCl_3 / $SOCl_2$.)

When ethylene glycol is treated with HI or P/I_2, 1,2 – diiodoethane is first formed which decomposes to give ethene.



ethane - 1,2 - diol 1,2 - diiodoethane ethene

On heating with conc HNO_3 in the presence of Con. H_2SO_4, ethylene glycol forms dinitroglycol.

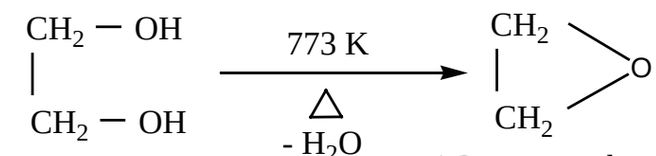


ethane - 1,2 - diol 1,2 - dinitroxyethene (dinitroglycol)

Dehydration reaction

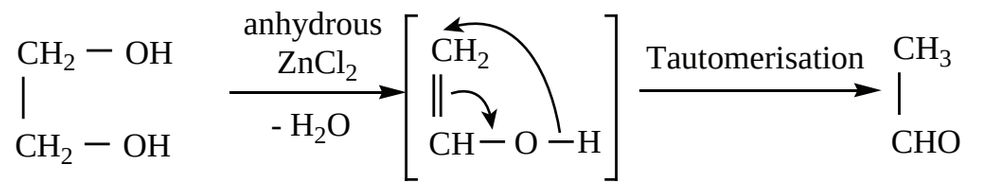
Ethyleneglycol undergoes dehydration reaction under different conditions to form different products.

1. When heated to 773K, it forms epoxides.



ethane - 1,2 - diol 1,2 - epoxyethane (Oxirane)

2. When heated with dilute sulphuric acid (or) anhydrous $ZnCl_2$ under pressure in a sealed tube, it gives acetaldehyde.



ethane -1,2 - diol ethenol (Vinylalcohol) Ethanal

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